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Appliance water filter system having memory and authentication functionality

Abstract

A water filter system for an appliance includes a filter head assembly, a first appliance control unit positioned proximate the filter head assembly and a water filter having a memory storage unit. The water filter is selectively placed in communication with the first appliance control unit to define a filter use state. The filter use state is further defined by a first potential fluid-flow rate through the water filter and a predetermined lifetime fluid filtration amount. A data transfer mechanism defines a second data-transfer protocol between the memory storage unit and the first appliance control unit when an actual fluid filtration amount of the water filter reaches the predetermined lifetime fluid filtration amount. The second data transfer protocol defines a notification state defined by a second potential fluid flow rate, the second potential fluid flow rate being less than the first potential fluid flow rate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 15/237,351, filed on Aug. 15, 2016, now U.S. Pat. No. 10,406,461, entitled APPLIANCE WATER FILTER SYSTEM HAVING MEMORY AND AUTHENTICATION FUNCTIONALITY. U.S. application Ser. No. 15/237,351 claims priority to Provisional Application Ser. No. 62/205,930, filed on Aug. 17, 2015, entitled APPLIANCE WATER FILTER SYSTEM HAVING MEMORY AND AUTHENTICATION FUNCTIONALITY, the disclosures of which are hereby incorporated herein by reference in their entirety.

FIELD OF THE DEVICE

(1) The device is in the field of water filtration systems, and more specifically, an appliance water filtration system having a memory for recording and communicating consumable status information and for providing authentication tags for use in conjunction with the appliance.

SUMMARY OF THE DISCLOSURE

(2) In at least one aspect, a water filter system for an appliance includes a filter head assembly, wherein a first appliance control unit is positioned proximate the filter head assembly. A water filter includes a memory storage unit wherein the water filter is selectively placed in communication with the first appliance control unit to define a filter use state. The filter use state is further defined by a first potential fluid-flow rate through the water filter and a predetermined lifetime fluid filtration amount. A data transfer mechanism is in communication with the memory storage unit and the first appliance control unit. The data transfer mechanism provides for a first data-transfer protocol between the memory storage unit and the first appliance control unit that further defines the filter use state. The data transfer mechanism also defines a second data-transfer protocol between the memory storage unit and the first appliance control unit when an actual fluid filtration amount of the water filter reaches the predetermined lifetime fluid filtration amount. The second data transfer protocol defines a notification state defined by a second potential fluid flow rate, the second potential fluid flow rate being less than the first potential fluid flow rate.

(3) In at least another aspect, a fluid filtration device for an appliance includes a water filter having a memory storage unit wherein the water filter is configured to be placed in selective communication with a first appliance control unit to define a unique filter identification tag. The unique filter identification tag defines a first fluid flow rate of a water filtration system of a first appliance. A data transfer mechanism is at least partially defined by the memory storage unit. The memory storage unit is placed in communication with the first appliance control unit. The data transfer mechanism, through cooperation of the memory storage unit and the first appliance control unit, defines a unique filter identification tag. The unique filter identification tag is at least partially stored within the memory storage unit, wherein after the unique filter identification tag is defined, the memory storage unit is free of cooperatively defining the unique filter identification tag with a second appliance control unit, wherein cooperation of the memory storage unit and the second appliance control unit defines a generic filter identification tag. The generic filter identification tag is configured to define a second fluid flow rate of a second water filtration system of a second appliance. The second fluid flow rate is less than the first fluid flow rate.

(4) In at least another aspect, a method for installing and replacing a water filter for an appliance includes providing a water filter having a memory storage unit, where the water filter is configured for selective communication with a first appliance control unit to define a filter use state. The filter use state is further defined by a first potential fluid-flow rate through the water filter and a predetermined lifetime fluid filtration amount. According to the method, the water filter is installed in communication with the first appliance control unit to place the memory storage unit in communication with the first appliance control unit, wherein a data transfer mechanism is defined by communication between the memory storage unit and the first appliance control unit, and where the data transfer mechanism defines a first data transfer protocol that further defines the filter use state. Fluid is selectively provided through the water filter at the first potential fluid-flow rate until

the actual fluid filtration amount of the water filter reaches the predetermined lifetime fluid filtration amount. A second data transfer protocol is defined through the data transfer mechanism when an actual fluid filtration amount of the water filter reaches the predetermined lifetime fluid filtration amount. The second data transfer protocol defines a notification state, and the second potential fluid flow rate is less than the first potential fluid flow rate. Next, the water filter is removed from communication with the first appliance control unit. The memory storage unit maintains the second data transfer protocol and is free of subsequently defining the first data transfer protocol. A replacement water filter has a replacement memory storage unit. The replacement water filter is configured for selective communication with the first appliance control unit to define the filter use state that is defined by the first potential fluid-flow rate through the replacement water filter and the predetermined lifetime fluid filtration amount. The replacement water filter is installed in communication with the first appliance control unit to place the replacement memory storage unit in communication with the first appliance control unit. The data transfer mechanism is typically defined by communication between the replacement memory storage unit and the first appliance control unit, and wherein the data transfer mechanism defines the first data transfer protocol that further defines the filter use state. The fluid is selectively provided through the replacement water filter at the first potential fluid-flow rate until the actual fluid filtration amount of the replacement water filter reaches the predetermined lifetime fluid filtration amount.

(5) In at least another aspect, a method for providing an automatic water filter exchange program for consumers includes providing a water filter having a memory storage unit, whereby the water filter is configured for selective communication with a consumer's cell phone NFC reader once installed in the refrigerator. This will take the consumer to a web site where they can register and sign up for automatic filter replenishment with or without filter analysis. When the consumer signs up for analysis, they will receive a return post paid container to return the used water filter. Once returned, the water filter NFC tag is read and water testing results provided to the consumer.

(6) Another aspect includes a fluid filtration combination system that includes: a fluid filtration device having: a water filter having a memory storage unit wherein the water filter is configured to be placed in selective communication with a first appliance control unit to define a unique filter identification tag. The unique filter identification tag defines a first fluid flow rate of a water filtration system of a first appliance. The system further includes a data transfer mechanism at least partially defined by the memory storage unit such that, when the memory storage unit is placed in communication with the first appliance control unit, the data transfer mechanism, through cooperation of the memory storage unit and the first appliance control unit, defines a unique filter identification tag. The unique filter identification tag is at least partially stored within the memory storage unit. After the unique filter identification tag is defined, the memory storage unit may be free of (prevented from) cooperatively defining the unique filter identification tag with a second appliance control unit. Additionally, the water filter may be further configured to be placed in communication with a mobile computing device that works in conjunction with the water filter and the memory storage unit of the water filter via a wireless data transfer device/system that typically includes an RFID reader associated with and typically engaged with or integrating into the mobile computing device such that the mobile computing device receives information about the water filter and optionally information stored in the memory storage unit of the water filter. It is also possible, but not necessary, as in other embodiments that cooperation of the memory storage unit and the second appliance control unit defines a generic filter identification tag, the generic filter identification tag may be configured to define a second fluid flow rate of a second water filtration system of a second appliance, the second fluid flow rate being less than the first fluid flow rate. The mobile computing device may be in wireless communication such that an automatic replacement of a water filter may be initiated and fulfilled. Additionally, the communication may prompt delivery of a package that receives the water filter when it is in an expired or used state. The package is

typically pre-addressed to a testing facility and/or recycling facility to test and or recycle the used filter. The testing information, if conducted may be reported to the user via the mobile computing device and a replacement filter with filtering features and capabilities tailored to the water filtering needs of the user may be provided to the user thereafter as well.

(7) Another aspect of the present disclosure includes a method of authenticating a water filter and/or communicating information about a water filter to a user via a user interface comprising the following steps: providing: a filter water filter system for an appliance, the water filter system characterized by: a filter head assembly; a first appliance control unit; a water filter having a memory storage unit engaged with the water filter; and a data transfer mechanism in communication with the memory storage unit and the first appliance control unit; positioning the water filter such that the memory storage unit is in communication with the first appliance control unit; communicating an initial startup protocol to the first appliance control unit after the water filter is initially engaged with filter head assembly and is in a first filter use state where water flows at an initial water flow rate to the user; communicating a subsequent data-transfer protocol between the memory storage unit and the first appliance control unit that overwrites the initial startup protocol; and providing information about the water filter to a user.

(8) Another aspect of the present disclosure includes a system for communicating information about a water filter to a user a filter water filter system for an appliance, the water filter system characterized by: a filter head assembly; a first appliance control unit; a water filter having a memory storage unit engaged with the water filter where the water filter is engaged with the appliance in a manner to allow untreated water to enter the water filter and treated water to be provided to the user via the appliance; and a data transfer mechanism that includes the memory storage unit and the first appliance control unit in signal communication with one another. The system periodically reads and writes information to the water filter's memory storage unit and the memory storage unit includes data that may be used to display one or more of the following pieces of status or history information about the water filter to a user via a user interface: the time period of use of the water filter; the volume of water that has flowed through the filter; the percentage of use or percentage of remaining useful life of the filter; the model of the filter; and the model of the appliance to which the filter was initially engaged.

(9) These and other features, advantages, and objects of the present disclosure will be further understood and appreciated by those skilled in the art by reference to the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing summary, as well as the following detailed description of the device, will be better understood when read in conjunction with the appended drawings. For the purpose of illustrating the device, there are shown in the drawings, certain embodiment(s) which are presently preferred. It should be understood, however, that the device is not limited to the precise arrangements and instrumentalities shown. Drawings are not necessary to scale. Certain features of the device may be exaggerated in scale or shown in schematic form in the interest of clarity and conciseness.

(2) FIG. 1 is a schematic, cross-sectional view of an appliance incorporating a water filter having an aspect of a memory storage unit for defining a data transfer mechanism;

(3) FIG. 2 is an enlarged schematic view of the water filtration system for the appliance of FIG. 1 showing the water filter engaged within a filter head assembly of the appliance;

(4) FIG. 3 is a schematic view of the water filter system showing the water filter operating during the filter use state;

- (5) FIG. 4 is a schematic view of the water filter system of FIG. 2 illustrating the water filter during the notification state;
- (6) FIG. 5 is a schematic view of the water filter system illustrating an exemplary process for the water filter being removed and the replacement water filter being installed within the filter head assembly;
- (7) FIG. 6 is a schematic diagram illustrating first and second appliances, each incorporating an aspect of the water filter system and illustrating sequential installation of a water filter in each of the first and second appliances;
- (8) FIG. 7 is a schematic diagram illustrating a non-authentication state defined between a first appliance control unit and a non-conforming water filter;
- (9) FIG. 8 is a schematic flow diagram illustrating a method for installing a water filter for an appliance incorporating an aspect of the water filter system; and
- (10) FIG. 9 is a schematic flow diagram illustrating a method for verifying the state of a water filter installed within a water filter system;
- (11) FIG. 10 is a schematic diagram illustrating an exemplary encryption/decryption system for the water filter system;
- (12) FIG. 11 is a schematic diagram illustrating an alternate aspect of an exemplary encryption/decryption system for the water filter system; and
- (13) FIG. 12 is a front schematic view of an appliance according to the present disclosure showing various exemplary locations where a user's mobile computing device or other reader may be placed to receive and transfer information about the water filter spaced within the appliance without the filter having to be removed from the appliance.

DETAILED DESCRIPTION

- (14) Before the subject device is described further, it is to be understood that the device is not limited to the particular embodiments of the device described below, as variations of the particular embodiments may be made and still fall within the scope of the appended claims. It is also to be understood that the terminology employed is for the purpose of describing particular embodiments or aspects of embodiments, and is not intended to be limiting. Instead, the scope of the claimed invention will be established by the appended claims.
- (15) Where a range of values is provided, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limit of that range, and any other stated or intervening value in that stated range, is encompassed within the device. The upper and lower limits of these smaller ranges may independently be included in the smaller ranges, and are also encompassed within the device, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the device.
- (16) In this specification and the appended claims, the singular forms “a,” “an” and “the” include plural reference unless the context clearly dictates otherwise.
- (17) Referring to the various aspects of the device as exemplified in FIGS. 1-6, reference numeral **10** generally refers to a water filter system for an appliance **12**. The water filter system **10** can include a filter head assembly **14** disposed within a portion of the appliance **12** and a first appliance control unit **16**, which is typically a microprocessor based control, with memory conducting arithmetic/logic operations and driving input and output devices to respond to a program's instructions. It directs the operation of the other appliance units by providing timing and operational control signals. Typically, the first appliance control unit is positioned within the appliance often proximate the filter head assembly **14**. The water filter system **10** also includes a water filter **18** having a memory storage unit **20**. The water filter **18** can be selectively placed in communication with the first appliance control unit **16** to define a filter use state **22**. The filter use state **22** is further defined by a first potential fluid flow rate **24** through the water filter **18** as well as

a predetermined useful lifetime fluid filtration amount **26** of water that passes through the water filter **18**. The first potential fluid flow rate **24** defines a maximum flow rate of water that can be passed through the water filter **18** over a period of time, for example, at least during the pendency of the filter use state **22**, typically where the filter is engaged with a refrigerator or other host appliance that dispenses, uses or dispenses and uses filtered water. The water to be filtered is typically water received by the appliance from a municipal water source or a well water source. Additionally, the predetermined lifetime fluid filtration amount **26** (See FIGS. 4-5) designates a total amount of water that can be passed through the water filter **18**, typically at the first potential fluid flow rate **24** during an overall useful life period of the water filter **18** indicative of the filter use state **22**.

(18) The water filter system **10** can also include a data transfer mechanism **28** that is in signal communication with the memory storage unit **20** and the first appliance control unit **16**. In this manner, the data transfer mechanism **28** provides for a first data-transfer protocol **30** between the memory storage unit **20** and the first appliance control unit **16** that is, typically, indicative of the filter use state **22**. According to various aspects of the device, the data transfer mechanism **28** can also provide a second data-transfer protocol **32** between the memory storage unit **20** and the first appliance control unit **16** that can be activated when an actual fluid filtration amount **34** of the water filter **18** reaches the predetermined lifetime fluid filtration amount **26**. It is contemplated that the second data transfer protocol **32** can be indicative of the notification state **36**. Once the water filter **18** reaches the notification state **36**, the water filter **18** may optionally change the water flow rate dispensed to the user, typically by slowing the rate of unfiltered water flowing through the water filter **18**. When this is done, water flows at a second potential fluid flow rate **38**. The second potential fluid flow rate **38**, according to the various embodiments, may be less than the first potential fluid flow rate **24**.

(19) Referring again to FIGS. 1-5, it is contemplated that the data transfer mechanism **28** can include the memory storage unit **20** and the first appliance control unit **16** in communication with one another. The two components may be physically engaged or connected. It is also contemplated, as specifically disclosed herein, that the components can be engaged and free of physical contact with one another when the water filter **18** is in the filter use state **22**. Accordingly, wireless communication between the memory storage unit **20** and the first appliance control unit **16** may be achieved through the use of various wireless communication methods. These wireless communication methods can include, but are not limited to, radio frequency identification (RFID), Wi-Fi, Bluetooth, near field communication (NFC), combinations thereof, and other similar wireless communication methods. As mentioned previously, it is also contemplated that an at least partial physical attachment communication between the memory storage unit **20** of the water filter **18** and the first appliance control unit **16** can be implemented. Communication between the memory storage unit **20** and at least one appliance control unit, or a plurality of control units, typically the first appliance control unit **16**, occurs when the water filter **18** is in the filter use state **22**. Such physical and/or wireless connection can implement any one of various communication signals, tags or other network communications between the memory storage unit **20** and the first appliance control unit **16**. It is also contemplated that the overall data transfer mechanism **28** can include both wireless and physically connected portions between the memory storage unit **20** and the first appliance control unit **16** when the water filter **18** is in the filter use state **22**.

(20) Referring again to FIGS. 1-7, the data transfer mechanism **28** defined between the memory storage unit **20** and the first appliance control unit **16** could be used to transfer various authentication tags as well as status data related to the use of the water filter **18** before, during and after the filter use state **22**. It is contemplated that the authentication tags can include various protocols that can include, but are not limited to, an initial start-up protocol **50**, the first data-transfer protocol, the second or a subsequent data-transfer protocol, a non-authentication protocol **52**, and other various protocols that will be described more fully below.

(21) According to the various embodiments, as exemplified in FIGS. 1-7, the initial start-up protocol **50** can be a set of encrypted instructions saved within the memory storage unit **20** of the water filter **18**. These encrypted instructions that define the initial start-up protocol **50** can be saved within the memory storage unit **20** during the process of manufacturing the water filter **18**. The initial start-up protocol **50** is configured to be communicated to an appropriate first appliance control unit **16** when the water filter **18** is first installed into a first appliance **60**. The initial start-up protocol **50** communicates an authentication signal **62** to the first appliance control unit **16**, where the first appliance control unit **16** decrypts the encrypted signal and compares the authentication signal **62** against an authentication identifier **64** saved within the first appliance control unit **16**. It is contemplated that the authentication identifier **64** disposed within the first appliance control unit **16** can also be encrypted. Accordingly, the authentication signal **62** provided by the initial start-up protocol **50** and the authentication identifier **64** disposed within the first appliance control unit **16** must include the encryption code such that the information can be exchanged according to a “hand shake” during the transfer of the initial start-up protocol **50** between the memory storage unit **20** to the first appliance control unit **16**.

(22) Once the authentication signal **62** and the authentication identifier **64** are decrypted, the authentication signal **62** and the authentication identifier **64** are compared, typically within and/or by the appliance control unit **16**, in order to authenticate the water filter **18** as a licensed and/or compatible water filter **18** for use in connection with the appliance **12** that the water filter **18** is engaged with. In order to authenticate the water filter **18**, various information exchanged between the memory storage unit **20** and the first appliance control unit **16** must match in order for the water filter **18** to be authenticated to initiate the filter use state **22** and the first data-transfer protocol. Such information can include, but is not limited to, one or a plurality of the following: a universal identifier containing a manufacturer ID, a non-stop forwarding (NSF) system certification number that can be unique to a particular manufacturer for a given water filter **18** and appliance **12** combination, the model of a particular water filter **18**, the serial number of a particular water filter **18**, the model of an appliance **12**, the serial number of an appliance **12**, and the various model number and serial number information written upon an authentication memory for purposes of “hand shake” communications between the memory storage unit **20** and the first appliance control unit **16**.

(23) According to various embodiments, as exemplified in FIGS. 1-11, various software packet coding can be transferred between the first appliance control unit **16** and the memory storage unit **20** to activate a second on-going authentication process. In this manner, the first data-transfer protocol, an authentication state for the water filter **18**, a read-write counter disposed within the first appliance control unit **16** and/or the memory storage unit **20**, combinations of the above, and/or other similar authentication elements can be electronically encrypted or otherwise encoded and used as part of the water filter authentication process to prevent unauthorized access to data stored on and communicated between the memory storage unit **20** and the first appliance control unit **16**. It is also contemplated that various status information regarding the water filter system **10** can be transferred between the memory storage unit **20** and the first appliance control unit **16**. Such status information can include, but is not limited to, fluid flow rates through the water filter **18**, the amount of fluid passed through the water filter **18**, the volume of water passing through the water filter **18**, end-of-life data, water filter expiration information, and other similar status information.

(24) Referring again to FIGS. 1-7, it is contemplated that the initial start-up protocol **50** saved within the memory storage unit **20** and/or the first appliance control unit **16** can be operated only once during the life cycle of the water filter **18** at the initial installation of the water filter **18** into an appropriate first appliance **60**. Accordingly, once the initial start-up protocol **50** is operated to completion and the water filter **18** is authenticated to initiate the first data-transfer protocol for operating the data transfer mechanism **28** defined between the memory storage unit **20** and the first appliance control unit **16**, the initial start-up protocol **50** can be overwritten. Accordingly, typically,

the initial start-up protocol **50** cannot be run again when the water filter **18** is disposed within a second appliance **90**. Accordingly, the overwriting of the initial start-up protocol **50** can prevent use of a single filter within multiple appliances **12**, where a filter may be attempted to be reused beyond the useful life of the water filter **18**. It is contemplated that a recycling processes **148** can be undertaken to provide for proper and authorized reuse of a water filter **18**. Such recycling processes **148** will be described more fully below.

(25) Referring again to FIGS. **1-7**, the use of the initial start-up protocol **50** can also be implemented to prevent use of nonconforming water filters **80** within a particular appliance **12**. As discussed above, the initial start-up protocol **50** can include, but is not limited to, an initial authentication “hand shake” that typically uses a combination of unique and proprietary encrypted passwords and/or proprietary microchip ID numbers. It is also contemplated that the initial start-up protocol **50** can implement a secondary “hand shake” between the first appliance control unit **16** and the memory storage unit **20** of the water filter **18** that also uses proprietary and encrypted certification numbers, encrypted proprietary, symmetrical or asymmetrical memory; security micro-control units; rolling serial numbers of the various matching water filters **18**; combinations thereof; and other security measures. It is further contemplated that the initial start-up protocol **50** can result in exchange of filter usage data as well as appliance **12** status data between the memory storage unit **20** and the first appliance control unit **16**. While this typically occurs at the startup for authentication, communication between the filter and the device that is reading the filter's tag (appliance or mobile device typically as will be described more below) may occur near the end of the filter's life based on estimated life, which is derived from usage time or totaled flow count. It may also occur periodically based on usage count or calendar date or externally sent subscription “ping” from a remote server(s), which would typically be owned or controlled by the manufacturer of the appliance or their representative. The information might also be read based on the occurrence of an event such as a power outage or upon filter removal and reinsertion. Finally, it is also contemplated that communication between the reading device and the water filter may be based upon a sensed event such as a change in water pressure, flow rate, temperature of water or ambient air or within the appliance, contaminant content within the untreated water to be treated, or a communication about water quality (a water quality alert) issued by a governmental authority, which would typically be a local utility or state or federal agency.

(26) According to the various embodiments, once the initial start-up protocol **50** is complete and the water filter **18** is authenticated for use within the first appliance **60**, the first data transfer protocol **30** can also include the use of various encryption technology for preventing unauthorized access to the authentication tags utilized during operation of the first data transfer protocol **30**. It is contemplated that through the use of these encryption and authentication mechanisms, the ability of an unauthorized user to decipher the various encryption and authentication tags can be made more difficult, thereby preventing or deterring use of nonconforming water filters **80**, which may not function as envisioned by the appliance **12** maker or may damage the appliance **12** from being used within various appliances **12**.

(27) Communication between the water filter and the appliance or another device, including a mobile computing device, may typically occur in one of two ways. First, the appliance may have a built in or be connected to a wireless computer network via a Wi-Fi connection or other wireless connection or via a wired connection such that it has the capability of communicating with the internet or an individual remote server or set of servers via a global communication network of computer servers. In this manner, the water filter may prompt communication with the user via the network to a user's computing device or mobile computing device causing information to be displayed to the user via a native mobile software application or via a website that displays information received from the refrigerator. Alternatively, the manufacturer or other provider may send a signal to the appliance and thereby to the water filter and receive information from the appliance about the filter use life and other information. Typically this would only be done after

receiving authorization from the user to do so such as in a user agreement via a mobile application or via another authorization method. Essentially, such a system would allow the refrigerator to communicate with the user at predetermined intervals or only when the filter is nearing the end of its useful life and/or at the end of its useful life. The notifications could be based on water quality or another event driven scenario that would potentially be measured by the refrigerator, the filter, or an outside data source from the individual users such as news and public utility warnings that might occur in a given geographic location or within a predetermined distance of the user/refrigerator. The other data could be water quality data received from testing done on other filters used in other appliances near or within a predetermined distance of the user's appliance as discussed herein. The filter and appliance communication could also cause the mobile application to display various notifications to the user via the mobile computer without the mobile application being open through notifications and other warnings. The notifications may further be communicated from a mobile application or via another route to a second mobile computing device or multiple other fixed or mobile computing devices such as a networked watch device configured to be in wireless communication with the device or the network directly via a wireless technology for exchanging data over short distances from fixed and mobile devices, most typically via a near field communication (NFC) specification, and building personal area networks (PANS), or other wireless communication method. The NFC tag on the filter typically combines an NFC Radio and an MCU (micro-control unit) on a single board to provide the NFC physical interface. The NFC tag communicates wirelessly with the NFC Module of the appliance. The NFC Module connects to Appliance communications/power bus (WIN) to communicate to the appliance's Appliance Control Unit (ACU) and Human Machine Interface (HMI). The Water Filter NFC tag Antenna will generally be sized to match the NFC Module Printed Circuit Board Antenna—25.91 mm×43 mm and provide a read range of at least 20 mm. The NFC TAG will typically include some or all of the following information: model number; serial number; filter creation date; filter manufacturer and/or supplier; filter actions including prohibit filter reset and modification of the flow rate; time limit of use of the filter; water volume limit of use of the filter; an installation date of when the filter was first installed into an appliance; a timer count that tracks time increments, typically ten minute increments; the water volume used; the time percentage used; and the volume percentage use. The base time for all time calculations is typically seconds such that a 6 month time period is 15,724,800 seconds (182 days×24 hr/day×60 min/hr×60 sec/min), for example, but could conceivably be any time increment. The NFC TAG will also have a shelf life of at least 10 years of life and an operational life of at least about 2 years.

(28) A second manner in which the filter may communicate is generally centered around ways the filter can communicate information to the user when the refrigerator itself does not have a wired or wireless connection to a network or even if the refrigerator does have a connection as described above. This further communication functionality allows the water filters incorporating an RFID communication tag or other communication tag having the functionalities described herein to provide the same information and functionality to users of appliances without a networked connection. Many refrigerators may not have a network connection and a RFID or other communication tag reader incorporated within the appliance. Basically, the systems of the present disclosure will work with a networked appliance but also will work with an external computing device, typically a mobile computing device such as a tablet computer or mobile cellular and Wi-Fi enabled phone, in particular, such devices that have an RFID reading device or other NFC (near field communication) reader. These devices may be placed within close proximity to the water filter's RFID chip, typically within about one inch or less and the device may read the information instead of or in addition to the appliance reader. The device of the user may then be used to communicate with the remote server(s); convey information about the filter to the user; and also to undertake the other actions and benefits of the systems described herein such as registration of the appliance, reordering of a replacement filter, and recycling and testing of used filters. The website

and mobile application may also track the shipment of replacement filters with RFID capability. The systems may also track and convey untreated water quality that is treated by the filter over time based on historical testing done on the tested and optionally recycled water filters.

(29) In addition to the water filters with the RFID tag on the water filter being able to be removed and read or read by another device when removed from engagement from the appliance, it is also possible that one may be able to read and communicate with the tag on the water filter without removing the filter from an appliance using a computing device that is not incorporated into the appliance such as a smartphone or smartwatch (iPhone®, iPad®, or Android® device). The user may place the user's device within a close proximity, a distance sufficiently close to read information on the device that is typically about one inch or less, by placing the device on a "landing zone." The landing zone **500** (see FIG. **12**) is any region or location inside or outside of the appliance where a smart device may be positioned so as to improve potential for reading the filter RFID tag without interference of signal. The landing zone may be internal or external to any appliance structure. It may be located in the Human Machine interface, in a portion of the dispenser well or around the dispenser, in the lower grill, between the crisper drawers or on a top, side or bottom surface of a liner of the door or a main appliance body or other interior compartment interior access point where there compartment and door or main appliance body are not below freezing. Its main function is to indicate to a user where to place the device to read the information on the filter. The landing zone may be an easily accessible flat planar surface adjacent the filter RFID tag when the filter is engaged or it may have a specific geometry so as to cradle or physically dock the smart device. Conceivably, the refrigerator could contain or have a remote dock that contains a wired connection to the appliance to enable communication of information about the filter from the RFID tag on the filter through the microprocessor of the appliance and to the device via a wired connection such as through a Lightning® cable in the case of an Apple smart device or other similar tethering device that can convey data and also typically power to the mobile device. In this manner a user might charge the mobile device by docking the device with the appliance and also receive information from the appliance without the need for a wireless network connection. In such situations however, the appliance will typically need to have the RFID reader incorporated therein unless the dock is located proximate the filter such as being incorporated into the landing zone. It should be understood that the information displayed to the user may be either displayed through use of a separate mobile application that at least partially resides on the user's device or via a web site browser URL system that will display through a device's website browser information based on the URL request received by the user's device. Either method of displaying and conveying information may be used in the context of any of the aspects of the present disclosure such that where an application is referenced a URL web browser system may also be used and vice versa.

(30) Referring now to FIGS. **5** and **6**, after a water filter **18** has been installed within a first appliance **60** and the initial start-up protocol **50** has been performed, and the water filter **18** authenticated, the water filter **18** can be designed such that the initial start-up protocol **50** cannot be run again. By way of example, and not limitation, the initial start-up protocol **50** can be overwritten, or computer code can be included to prevent a second operation of the initial start-up protocol **50**. Accordingly, after the water filter **18** has been installed and authenticated within the first appliance **60**, an attempt to install the same water filter **18** into a second appliance **90** can result in the operation of a non-authentication protocol **52**, such that the water filter **18** cannot be authorized by the second appliance **90**. In such a circumstance, the filter use state **22** cannot be activated and the first data-transfer protocol cannot be run within the second appliance **90**.

(31) According to various alternate embodiments, it is contemplated that the initial start-up protocol **50** will only be written over when the water filter **18** has substantially passed or passed its useful life and is disposed in the notification state **36**. In such an embodiment, so long as the water filter **18** is in the filter use state **22**, the water filter **18** can be used in other corresponding appliances **12** and the initial start-up protocol **50** is capable of being executed to authenticate the water filter **18** in

each of the appliances **12**. During the filter use state **22** of the water filter **18**, status information regarding use of the water filter **18** may be continually recorded and updated within the memory storage unit **20** to continually define and redefine the current status information of the water filter **18**. This current or “real time” status information is transferred to the corresponding first appliance control unit **16** of each appliance **12** with which the water filter **18** has been authorized to work. In addition to current or “real time” status information, history and use information through any portion or all of the useful life of the water filter **18** may be transferred. The water filter **18** may be disposed in the notification state **36** when substantially all or all of the useful life of the water filter **18** has passed, whether the water filter **18** has been used in a single appliance **12** or multiple appliances **18**. Once the notification state **36** has been reached, it is contemplated that both the initial start-up protocol **50** and the first data transfer protocol **30** can be overwritten. Accordingly, once the water filter is in the notification state **36**, the water filter **18** may not be installed within another appliance **12**. Whether the water filter **18** is installed in a first appliance **60**, a second appliance **90**, or a subsequent appliance **12**, the user is accurately informed when a water filter **18** is or has become expired regardless of whether the water filter **18** was used in one or more appliances **12** previously.

(32) It is contemplated that the non-authentication protocol **52** can result in various modifications to the water filter system **10** and/or notifications to the user. Such modifications and notifications can include, but are not limited to, a diminished flow rate of fluid through the water filter **18**, deactivation of the water system of the appliance **12** such that water cannot be delivered through the water filter **18**, activation of a warning signal **100** associated with the appliance **12** alerting the user that the filter trying to be used is a non-conforming water filter **80**, combinations thereof, and other similar identifiers, which may or may not be communicated to the user of the second appliance **90**, but typically are communicated or are communicated with an appliance user. A nonconforming water filter **80** may be one that has an improper protocol, tag or other signal that does not match the information stored in the first appliance control unit **16**. A nonconforming water filter **80** may also be a water filter that has no memory storage unit **20** or is otherwise unable to transfer data.

(33) The use of the non-authentication protocol **52** can also be implemented to prevent unauthorized recycling of used water filters **18** that have been at least partially used in a first appliance **60**, and can also prevent unauthorized counterfeiting of water filters **18** that may not conform to the requirements necessary for use in a particular appliance **12**. Accordingly, the non-authentication protocol **52** can include a non-authentication signal **110** that is delivered from the memory storage unit **20** to the appliance control unit **16** of the second appliance **90**. This non-authentication signal **110** can be implemented when the authentication identifier **64** disposed within the appliance control unit of the second appliance **90** either does not receive the authentication signal **62** or receives a non-authentication signal **110**. As discussed above, when a water filter **18** has been previously used such that the initial start-up protocol **50** is overwritten, the authentication signal **62**, according to the various embodiments, may not be sent, as the initial start-up protocol **50** for providing an authentication signal **62** may no longer be present within the memory storage unit **20**.

(34) Referring again to FIGS. **1-7**, the first data transfer protocol **30** of the data transfer mechanism **28** of the first appliance **12** that is performed when the water filter **18** is engaged with the appliance **12** can operate until such time as the actual fluid filtration amount **34** of the water filter **18** reaches the predetermined lifetime fluid filtration amount **26** saved within a portion of the water filter system **10**. At this point, the data transfer mechanism **28** can switch from the first data-transfer protocol to the second data-transfer protocol that is operated during the notification state **36** of the water filter **18**. As discussed above, the notification state **36** of the water filter **18** is defined by the water filter **18** having filtered an amount of water equal to, or substantially equal to, the predetermined lifetime fluid filtration amount **26**. Additionally, due to the single-use nature of the

initial start-up protocol **50**, it is contemplated that this amount of water is filtered during use of the water filter **18** within only the first appliance **60**. Once the second data-transfer protocol is initiated, the second data transfer protocol **32** can define various changes to the water filter system **10** and the first appliance **60** in general. Such changes can include, but are not limited to, activation of the second potential fluid flow rate **38**, where the second potential fluid flow rate **38** can be less than the first potential fluid flow rate **24**. This changed fluid flow or other communication to the user of the appliance **12** would indicate to the user that the water filter **18** has exceeded or is about to exceed its effective useful life.

(35) By way of example, and not limitation, the second potential fluid flow rate **38** can be, at most, 0.5 times the first potential fluid flow rate **24**. Alternatively, the second potential fluid flow rate **38** can be defined by a sequentially decreasing flow rate that generally decreases from the first potential fluid flow rate **24** and downward to a level that can be between 0.5 times the first potential fluid flow rate **24** to zero fluid flow through the water filter **18** and/or the fluid delivery system **122** of the appliance **12**.

(36) It is also contemplated that when the second data transfer protocol **32** is initiated, the second potential fluid flow rate **38** can be immediately changed to zero fluid flow through the water filter **18** and/or the fluid delivery system **122** of the appliance **12**. This change in the potential fluid flow rate when the water filter **18** changes from the filter use state **22** to the notification state **36** can be performed by a mechanism disposed within the water filter **18**, a mechanism disposed within the filter head assembly **14** of the appliance **12**, a valve **120** disposed within the fluid delivery system **122** of the first appliance **60**, combinations thereof, and other various mechanisms that can alter the amount of water delivered to and through the water filter **18** during the notification state **36** of the water filter **18** such that substantially all of the effective useful life of the water filter **18** has passed.

(37) Referring again to FIGS. **1-7**, once the water filter **18** is disposed within the notification state **36**, the expired status of the water filter **18** being in the notification state **36** may be saved within the memory storage unit **20** of the water filter **18**. Accordingly, subsequent use of the water filter **18** might be limited to the second potential fluid flow rate **38** defined by the notification state **36** of the water filter **18**. According to the various embodiments, it is contemplated that once the notification state **36** of the water filter **18** is reached, such that substantially all or all of the effective useful life of the water filter **18** is passed, data related to the filter use state **22** can be overwritten, deleted, or otherwise prevented from being subsequently used. In this manner, once the water filter **18** reaches the notification state **36**, only the second potential fluid flow rate **38** can be implemented during use of the water filter **18** in the first appliance **60** or any other conforming appliance **12**. The use of the notification state **36** can prevent and/or deter use of the water filter **18** beyond the effective life of the filtration media **130** disposed within the water filter **18**. After a certain amount of water has been delivered through the water filter **18** corresponding to the predetermined lifetime fluid filtration amount **26**, it is possible that the filtration media **130** can be less effective such that the water filter **18** may be less able to remove various materials from water delivered through the water filter **18**. Accordingly, the use of the notification state **36** and the second potential fluid flow rate **38** can communicate to the user that the filtration media **130** has reached or substantially reached its life cycle and should be replaced with a replacement water filter **140**.

(38) According to the various embodiments, once the water filter **18** has reached the notification state **36**, and substantially all or all of the useful life of the water filter **18** has been reached or exceeded, it is contemplated that a recycling process **148** can be implemented to recondition the water filter **18** for authorized reuse in another appliance **12** such that the initial start-up protocol **50** can be executed after recycling. Such a recycling process **148** can include replacing the memory storage unit **20** along with the filtration media **130**. The replacement memory storage unit **150** can include the data for initiating the initial start-up protocol **50** when the water filter **18** is installed within an appliance **12**. It is also contemplated that the recycling process **148** can include

reprogramming the memory storage unit **20** to erase all of the previously recorded status information and include data for initiating the initial start-up protocol **50**. According to the various embodiments, the recycling process **148** can include a system for monitoring the number of times that a particular water filter **18** has been recycled. This system can also include the storage of data related to overall use of the water filter **18**. It is contemplated that such overall use data can be sequestered from the appliance **12** and only utilized in the recycling process **148** to determine if recycling or disposal is appropriate. The overall use data can be stored in a separate memory unit **160** positioned within the water filter **18**.

(39) Referring again to FIGS. **1-7**, the water filter system **10** can include a fluid filtration device **180** for the appliance **12**, where the fluid filtration device **180** includes the water filter **18** having a memory storage unit **20** disposed thereon. As discussed above, the water filter **18** is configured to be placed in selective communication with the first appliance control unit **16** to define a unique filter identification tag **182** that can correspond to the first data transfer protocol **30**. The unique filter identification tag **182** can define the first potential fluid flow rate **24** of the water filter system **10** for a first appliance **60**. The fluid filtration device **180** can also include the data transfer mechanism **28** that is at least partially defined by the memory storage unit **20** being placed in communication with the first appliance control unit **16**. In this manner, the data transfer mechanism **28**, through cooperation of the memory storage unit **20** and the first appliance control unit **16**, can define the unique filter identification tag **182**. It is contemplated that the unique filter identification tag **182** is at least partially stored within the memory storage unit **20**. After the unique filter identification tag **182** is defined, the memory storage unit **20** is subsequently unable to cooperatively define the unique filter identification tag **182** with a second appliance control unit **92** of a second appliance **90**. Accordingly, cooperation of the memory storage unit **20** and the second appliance control unit **92** can define a generic filter identification tag **190**, where the generic filter identification tag **190** is configured to define the second potential fluid flow rate **38** of a second water filter system **94** of the second appliance **90**. As discussed above, the second potential fluid flow rate **38** is less than the first potential fluid flow rate **24**. It is contemplated that the generic filter identification tag **190** can correspond to one or both of the non-authentication protocol **52** or the second data transfer protocol **32**, depending upon the configuration of the memory storage unit **20** of the water filter **18** and/or the first and second appliance control units **16**, **92**.

(40) Referring again to FIGS. **1-7**, the memory storage unit **20** can define the predetermined lifetime fluid filtration amount **26** of untreated water that can be delivered through the water filter **18** to effectively form/yield treated water. It is contemplated that the unique filter identification tag **182** that corresponds to the first data transfer protocol **30** can be operative until the actual fluid filtration amount **34** of the water filter **18** reaches the predetermined lifetime fluid filtration amount **26** of the water filter **18**. At this point, the memory storage unit **20** can cease to generate the unique filter identification tag **182**, or the data related to the unique filter identification tag **182** can be overwritten. The memory storage unit **20** can then generate one or more of the generic filter identification tag **190**, the non-authentication protocol **52**, or the second data transfer protocol **32**. Accordingly, when a user attempts to use the water filter **18** in the second appliance **90** and/or when the water filter **18** has exceeded its useful lifespan, the water filter system **10** is configured to change in some manner to communicate to the user that the water filter **18** is ready to be replaced with a new water filter **18** that can run the initial start-up protocol **50** and, subsequently, the first data transfer protocol **30**. It is contemplated that the unique filter identification tag **182** can include one or both of the initial start-up protocol **50** and the first data transfer protocol **30**.

(41) Referring again to FIGS. **1-7**, it is contemplated that the filter use state **22** of the water filter **18** can be defined by the memory storage unit **20** of the water filter **18** being spaced approximately one inch, or less, from the first appliance control unit **16**, such distances between the memory storage unit **20** and the first appliance control unit **16** can be one-half inch, one-quarter inch, or other similar distance. The close positioning between the memory storage unit **20** and the first

appliance control unit **16** can be used where the data transfer mechanism **28** is a radio frequency identification (RFID) mechanism. In embodiments where an RFID mechanism **200** is implemented, an RFID microchip can be disposed on a water filter **18** and a radio circuit module **204** can be disposed on the first appliance control unit **16** or otherwise within the appliance **12** at a location capable of reading a signal. Additionally or alternatively, a mobile computing device may be used as a radio circuit module/RFID reader as will be discussed further below. The RFID mechanism **200** can implement the use of electromagnetic fields to transfer data for purposes of automatically identifying and tracking various tags associated with the memory storage unit **20** and the first appliance control unit **16**. These tags, which can correspond to the various protocols and signals described above, can deliver electronically stored information between the memory storage unit **20** and the first appliance control unit **16** or potentially any other RFID reader. Also as discussed above, this information can be encrypted to prevent unauthorized access.

(42) In addition to the delivery of information, it is contemplated that the RFID microchip disposed on the water filter **18** can be powered by the radio energy transmitted by the radio circuit module **204** at least when the water filter **18** is in the filter use state **22**. Accordingly, the various tags, signals and protocols can be powered through the use of electromagnetic induction generated from the magnetic fields used to transfer the data via the RFID mechanism **200**. It is also contemplated that the memory storage unit **20** and the first appliance control unit **16** can include a dedicated power source disposed within the water filter **18** and the filter head assembly **14**, respectively. Additionally, it is contemplated that the RFID microchip **202** can be disposed within a portion of the water filter **18**. In such an embodiment, the RFID microchip **202** can be a printed microchip disposed on an exterior surface of the water filter **18**, on an interior surface of the water filter **18**, such as within an inner cavity of the water filter **18**, or some other surface of the water filter **18**. It is also contemplated that the RFID microchip **202** can be embedded within a portion of the water filter **18**, such as within one of the walls of the water filter **18** or proximate the filtration media **130** of the water filter **18**. The microchip **202** may be molded into a portion of the filter housing, for example. It is contemplated that the RFID microchip **202** may be embedded in a tamper-evident location such that if the RFID microchip **202** is accessed, or attempted to be accessed, by a user, the water filter **18** will be not engage the appliance **12** properly or, at a minimum, the tampering would be readily visible to a subsequent user.

(43) According to the various embodiments, it is contemplated that alternate data transfer mechanisms **28** can be implemented within a water filter system **10**. Such data transfer mechanisms **28** can include an antenna disposed proximate a portion of the memory storage device, such as near a microchip of the water filter **18**. In such an embodiment, the first appliance control unit **16** can be positioned proximate a main control housing of the first appliance **60**. It is contemplated that the main control housing can be positioned distal from the water filter **18**, such as at the main control unit controlling the various mechanical and electrical functions of the entire appliance **12**. Accordingly, the antenna can be in communication with the main control housing when the water filter **18** is in the filter use state, even though the water filter **18** may be positioned on a distal portion of the appliance **12**.

(44) According to the various embodiments, the disclosure includes providing: the data transfer mechanism **28**, the memory storage unit **20** and the first appliance control unit **16**; and/or the memory storage unit **20** and a user's mobile computing device that can cooperate to deliver information to a manufacturer concerning status information of the appliance **12** and/or the water filter system **10**. In such an embodiment, where the user of the appliance **12** authorizes the delivery of such information, the manufacturer can be notified when a nonconforming water filter **80**, such as a counterfeit water filter or a water filter **18** that is in the notification state **36** (has substantially or completely expired), is attempted to be used within the appliance **12**. It is possible for the manufacturer to then contact the user to inform them of how to obtain a conforming water filter **18** for use in the appliance **12**. Such communications can be delivered to the manufacturer through a

Wi-Fi communications network, other wireless communications network, a wired communications network, combinations thereof, or other similar communications network. Such data transfer to the manufacturer can also alert the manufacturer to any unauthorized attempts to access encrypted information contained within the water filter **18** and/or the appliance **12**. Such communication may also be made through a mobile application running on the user's mobile computing device such as a smartphone (typically with cellular and/or Wi-Fi capability), tablet or laptop computer systems or through a website (URL) enable system accessed by the user via an Internet browser. Traditional desktop computer systems may also be used, but would be less likely to be used to communicate such information. The mobile computing devices, when used, will typically have a RFID reader mechanism incorporated therein.

(45) According to the various embodiments, it is contemplated that the data transfer mechanism **28** can be in signal communication with a portable computing device for communicating data from the data transfer mechanism to a separate location. Such a portable computing device can include smart phones, tablets, personal computers, wearable computing devices, or other similar devices having data processing and communication capability.

(46) Referring again to FIGS. **1-7**, the use of the initial start-up protocol **50** can also be implemented to provide an automatic exchange program for consumers. A consumer concerned about the quality of his/her water will typically send water samples into independent water testing facilities. This often requires the consumer to purchase a sampling kit at a local store, obtain a water sample, fill out name and address information, mail the sample to the testing facility, and wait for the test results in the mail. This process is burdensome and time consuming. Further, the results are not usually digitally available and do not provide comparative test results from previous testings, nor do they provide recommendations as to how to filter the water more effectively. As already described above, the initial start-up protocol **50** can include, but is not limited to, an initial authentication "hand shake" that typically uses a combination of unique and proprietary encrypted passwords and/or proprietary microchip ID numbers. It is further contemplated that the initial start-up protocol **50** can result in exchange of filter data and a consumer's phone or other mobile device. The consumer or user may elect to enroll in the automatic exchange program and elect to have the filter analyzed or not upon expiration of the filter's useful life. If enrolled and upon the expiration of the useful life of the water filter **18**, the consumer will use the mobile phone (or other device) NFC reader to read the water filter **18**. The user may be taken to a particular display of information in a mobile application or be taken to a web site. At either type of data entry point, in order to enroll in the automatic exchange program either with or without analysis, the consumer typically registers their personal identifying information (i.e. name, address, telephone number)(with the appliance model and number automatically filled in) whereby the consumer can sign up for an automatic filter replenishment with or without analysis. If the user elects to have an analysis of the filter's use done, the consumer will receive a return post paid container in the mail to the registered address for the user that was provided, typically via an internet website, via a computer network, or via a mobile application similarly in communication with a computer network. Typically, the computer network is a remote computer network connected to the Internet. The consumer can then return the used water filter to the manufacturer for testing or to a testing facility, which is typically a testing facility designated by the manufacturer. Once the water filter has been shipped to the testing site and received by the testing facility, the filter is placed into a water test machine that will read the data from the NFC tag and the data in the memory storage unit **20** of the water filter **18**, create a water report, and send the results to the user's account where the report and information may be accessed by the consumer via the web site or the mobile application. Of course, the consumer may have the filter automatic replaced without any testing done to the filter.

(47) The data from the water filter NFC tag and/or memory storage unit can be used by the water testing facility to determine, for example, types of water that require different types of filtering and thereby offer or recommend to the consumer water filters that are more appropriate for the type of

water they have in their system(s) in their specific geographic location. The data from multiple consumers may also be aggregated and sorted by type of appliance, geographic location (zip code, street address, city, state/province, etc. and recommendations made to future first time filter users based on data from other nearby users. This might typically be done: at the time the appliance is registered through the website of the manufacturer of an appliance; at the time and location of purchase of an appliance; at the time the appliance is initially registered by the consumer through use of the mobile application or at a time after registration of the appliance, but before any filter that has been used in the consumer's appliance is actually tested; at any time the consumer desires, a filter fitting the generally filtering needs of the geographic area of future use may be ordered by the user so long as enough other consumers in the general geographic area have had their filters tested and analyzed.

(48) Referring now to FIGS. **1-9**, having described the various aspects of the water filter system **10** that incorporates the data transfer mechanism **28** defined by the memory storage unit **20** of the water filter **18** and the first appliance control unit **16** of a first appliance **60**, a method is also disclosed for installing and replacing a water filter **18** for an appliance **12**. Such a method **400** can include a step **402** of providing a water filter **18** having a memory storage unit **20**. As discussed above, the water filter **18** can be configured for selective communication with a first appliance control unit **16** to define the filter use state **22**. The filter use state **22** can be further defined by a first potential fluid flow rate **24** through the water filter **18** and the predetermined lifetime fluid filtration amount **26**. According to step **404** of the method **400**, the water filter **18** can be installed within the filter head assembly **14** of the appliance **12** to be in communication with the first appliance control unit **16**.

(49) Once the water filter **18** is installed, it is determined whether the initial startup protocol **50** is present on the water filter **18** (step **404**). Where the initial startup protocol **50** is not present, a non-authentication protocol **52** is run (step **406**) and the fluid delivery system **122** is modified (step **408**) in some manner to slow or stop the flow of water and/or to activate a warning signal **100**. The non-authentication protocol **52** may be activated where the initial start-up protocol **50** has previously been overwritten through previous use of the water filter **18**, or where the water filter **18** is in a non-conforming water filter **80**.

(50) If the initial start-up protocol **50** is present, it is run (step **410**) and the first appliance control unit **16** receives and decrypts the data and compares the data for authorization purposes (steps **412-416**). If there is no match, the non-authentication protocol **52** is run. If a match is found, the initial start-up protocol **50** is overwritten (step **418**) and the first data transfer protocol **30** is run (step **420**). This engagement places the memory storage unit **20** in communication with the first appliance control unit **16**, such that the data transfer mechanism **28** can be defined by communication between the memory storage unit **20** and the first appliance control unit **16**. In this manner, the data transfer mechanism **28** can define the first data transfer protocol **30** that further defines the filter use state.

(51) Once installed, fluid is selectively provided through the water filter **18** at the first potential fluid flow rate **24**. This first potential fluid flow rate **24** continues until the actual fluid filtration amount **34** of the water filter **18** reaches the predetermined lifetime fluid filtration amount **26** (step **422**). At this point, the first data transfer protocol **30** is overwritten (step **424**) and the second data transfer protocol **32** is defined through the data transfer mechanism **28** (step **426**). Accordingly, the second data transfer protocol **32** defines the notification state **36**. As discussed previously, the second potential fluid flow rate **38** can be less than the first potential fluid flow rate **24**. When the notification state **36** is defined, the user can be notified that the water filter **18** is ready to be replaced. At this time, the user can remove the water filter **18** from communication with the first appliance control unit **16**. After being removed from the first appliance control unit **16**, the memory storage unit **20** maintains the second data transfer protocol **32** and is free of subsequently defining the first data transfer protocol **30**. As discussed previously, it is contemplated that the second data

transfer protocol **32** can overwrite the first data transfer protocol **30** such that the first data transfer protocol **30** cannot be subsequently reinitiated when disposed within the same appliance **12** or a different appliance **12**.

(52) According to another aspect of method **400**, as exemplified in FIG. **9**, a replacement water filter **140** can be provided having a replacement memory storage unit **160** or a reprogrammed memory storage unit **20**. It is contemplated that the replacement water filter **140** is configured for selective communication with the first appliance control unit **16** to define the filter use state **22** as to the first appliance control unit **16** and the replacement memory storage unit **160**. The replacement water filter **140** can then be installed (step **450**). Once installed, the status of the water filter **18** is determined to ascertain whether the appliance **12** has already authorized the water filter **18** (step **402**). If already authenticated, the water filter **18** is not a replacement water filter **140** and it is determined whether the water filter **18** is in the filter use state **22** (step **454**) or the notification state **36** (step **456**). If in the notification state **36**, the second data transfer protocol **32** is continued (step **460**) and the second potential fluid flow rate **38** is used. If still in the filter use state **22**, the first data transfer protocol **30** is used to deliver water at the first potential fluid flow rate **24** (step **462**). The replacement water filter **140** can be placed in communication with the first appliance control unit **16** to place the replacement memory storage unit **160** in communication with the first appliance control unit **16**. In this manner, the data transfer mechanism **28** is again defined by communication between the replacement memory storage unit **160** and the first appliance control unit **16**. It is further contemplated that the data transfer mechanism **28** can be defined by the first data transfer protocol **30** that further defines the filter use state **22**. Once the replacement water filter **140** is installed, fluid is selectively provided through the replacement water filter **140** at the first potential fluid flow rate **24** until such time as the actual fluid filtration amount **34** of the replacement water filter **140** reaches the predetermined lifetime fluid filtration amount **26**. At this time, the various steps of replacing the water filter **18** can be repeated continuously through the life of the appliance **12**.

(53) It is important to note that the construction and arrangement of the elements of the device as shown in the exemplary embodiments is illustrative only. Although only a few embodiments of the present innovations have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements shown as multiple parts may be integrally formed, the operation of the interfaces may be reversed or otherwise varied, the length or width of the structures and/or members or connector or other elements of the system may be varied, the nature or number of adjustment positions provided between the elements may be varied. It should be noted that the elements and/or assemblies of the system may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present innovations. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the desired and other exemplary embodiments without departing from the spirit of the present innovations.

(54) It will be understood that any described processes or steps within described processes may be combined with other disclosed processes or steps to form structures within the scope of the present invention. The exemplary structures and processes disclosed herein are for illustrative purposes and are not to be construed as limiting.

(55) It is also to be understood that variations and modifications can be made on the aforementioned structures and methods without departing from the concepts of the present

invention, and further it is to be understood that such concepts are intended to be covered by the following claims unless these claims by their language expressly state otherwise.

Claims

1. A refrigeration appliance water filter comprising: a housing having filter media therein; a memory device disposed within or on the housing and wherein the memory device includes an initial startup protocol, a subsequent data transfer protocol that defines and updates a filter use state, and a non-authentication protocol stored therein and wherein the initial startup protocol comprises encrypted instructions used to authenticate the refrigeration appliance water filter that is configured to be executed a first time when the refrigeration appliance water filter is engaged with a filter head assembly of a refrigeration appliance; and wherein the initial startup protocol is overwritten such that it cannot be executed a second time after it is executed and a first fluid flow rate of filtered water is established by the refrigeration appliance water filter.
2. The refrigeration appliance water filter of claim 1, wherein the encrypted instructions are transmitted to a microprocessor based control within the refrigeration appliance using a wireless signal exchanged between the memory device and the microprocessor based control within the refrigeration appliance and wherein the initial startup protocol authenticates the refrigeration appliance water filter as a licensed and/or compatible water filter that remains effective at filtering water.
3. The refrigeration appliance water filter of claim 2, wherein the wireless signal is a RFID signal and wherein the encrypted instructions exchanged between the memory device of the refrigeration appliance water filter and the microprocessor based control within the refrigeration appliance comprises a unique filter identification tag stored in the memory device of the refrigeration appliance water filter and chosen from the group consisting of: a universal identifier containing a manufacturer ID, a non-stop forwarding (NSF) system certification number unique to a manufacturer for a given water filter and appliance combination, a model of the water filter, a serial number of the water filter, an appliance model, an appliance serial number, and combinations thereof.
4. The refrigeration appliance water filter of claim 3, wherein the memory device stores information regarding the refrigeration appliance water filter's use status within the memory device.
5. The refrigeration appliance water filter of claim 1, wherein the memory device stores information regarding the refrigeration appliance water filter's use status within the memory device.
6. The refrigeration appliance water filter of claim 2, wherein the encrypted instructions exchanged between the memory device of the refrigeration appliance water filter and the microprocessor based control within the refrigeration appliance comprises a unique filter identification tag stored in the memory device of the refrigeration appliance water filter and chosen from the group consisting of: a universal identifier containing a manufacturer ID, a non-stop forwarding (NSF) system certification number unique to a manufacturer for a given water filter and appliance combination, a model of the water filter, a serial number of the water filter, an appliance model, an appliance serial number, and combinations thereof.
7. The refrigeration appliance water filter of claim 3, wherein the water filter further comprises an RFID microchip disposed on the refrigeration appliance water filter that transmits the wireless signal and wherein the RFID microchip is powered by radio energy transmitted by a radio circuit module of the refrigeration appliance.
8. The refrigeration appliance water filter of claim 1, wherein the refrigeration appliance water filter is designed to filter water received by the refrigeration appliance of a refrigeration appliance water filter user from a water source within a given geographic area based upon a water quality of water received from a municipal water source or a well water source within the given geographic area of the refrigeration appliance water filter user.

9. The refrigeration appliance water filter of claim 8, wherein the water quality of water received from the municipal water source or the well water source is determined by testing of a plurality of used water filters from the given geographic area.

10. The refrigeration appliance water filter of claim 1, wherein the refrigeration appliance is connected with a water filter provider using a wired connection or a wireless connection to connect the refrigeration appliance with a server via a global communication network of computer servers and wherein refrigeration water filter status information is provided to a microprocessor based control within the refrigeration appliance and thereafter to a user via a mobile computing device accessed by the user.

11. The refrigeration appliance water filter of claim 1, wherein the initial startup protocol of the refrigeration appliance water filter is overwritten after the refrigeration appliance water filter has reached or exceeded substantially all or all of a useful life of the refrigeration appliance water filter through previous use of the refrigeration appliance water filter.

12. The refrigeration appliance water filter of claim 11, wherein the refrigeration appliance water filter provides filtered water at a first fluid rate after the refrigeration appliance water filter is authenticated and the refrigeration appliance water filter provides water at a second fluid flow rate after the refrigeration appliance water filter has reached or exceeded substantially all or all of the useful life through previous use of the refrigeration appliance water filter.

13. The refrigeration appliance of claim 12, wherein the refrigeration appliance water filter stops filtered water from flowing through the refrigeration appliance water filter after the refrigeration appliance water filter has reached or exceeded substantially all or all of the useful life through previous use of the refrigeration appliance water filter.

14. An appliance water filter comprising: a water filter housing having water filter media therein; a memory device disposed within or on the water filter housing and wherein the memory device includes an initial startup protocol and a data transfer protocol, wherein the initial startup protocol is wirelessly transmitted using a wireless signal to a microprocessor based control with memory within a refrigeration appliance and the initial startup protocol comprises an encrypted authentication tag that is wirelessly transmitted to the microprocessor based control with memory and used to authenticate the appliance water filter that is configured to be executed when the appliance water filter is engaged with the refrigeration appliance, and the data transfer protocol defines and updates a filter use state; and wherein the initial startup protocol is configured to be overwritten after it is executed such that it cannot be executed a second time.

15. The appliance water filter of claim 14, wherein the wireless signal is a RFID signal and wherein the initial startup protocol comprises encrypted instructions exchanged between the memory device of the appliance water filter and the microprocessor based control within the refrigeration appliance comprises a unique filter identification tag stored in the memory device of the appliance water filter and chosen from the group consisting of: a universal identifier containing a manufacturer ID, a non-stop forwarding (NSF) system certification number unique to a manufacturer for a given water filter and appliance combination, a model of a water filter, a serial number of the water filter, an appliance model, an appliance serial number, and combinations thereof.

16. The appliance water filter of claim 14, wherein the initial startup protocol is prevented from being performed more than one time and filtered water is no longer supplied to the refrigeration appliance from the appliance water filter when either (1) the initial startup protocol has been previously overwritten after substantially all or all of a useful life of the appliance water filter has been reached or exceeded through previous use of the appliance water filter or (2) where the appliance water filter is a non-conforming water filter that does not have an authentication signal that matches an authentication identifier of the microprocessor based control.

17. The appliance water filter of claim 16, wherein the refrigeration appliance is connected with a water filter provider using a wired connection or a wireless connection to connect the refrigeration appliance with a server via a global communication network of computer servers and wherein

appliance water filter status information is provided to the microprocessor based control within the refrigeration appliance and thereafter to a user via a mobile computing device accessed by the user.

18. An appliance water filter comprising: a water filter housing having water filter media therein; a memory device disposed within or on the water filter housing and wherein the memory device includes an initial startup protocol and a data transfer protocol, wherein the initial startup protocol is transmitted to a microprocessor based control with memory within a refrigeration appliance and the initial startup protocol comprises an encrypted authentication tag that is transmitted to the microprocessor based control with memory and used to authenticate the appliance water filter, wherein the initial startup protocol is configured to be executed when the appliance water filter is engaged with the refrigeration appliance, and the data transfer protocol defines and updates a filter use state; and wherein the initial startup protocol is overwritten after it is executed such that it cannot be executed a second time.

19. The appliance water filter of claim 18, wherein the appliance water filter is designed to filter water received by the refrigeration appliance from a water source within a given geographic area based upon a water quality of water received from a municipal water source or a well water source within the given geographic area of a user and wherein the water quality of water received from the municipal water source or the well water source is determined by testing of a plurality of used water filters from the given geographic area and wherein the refrigeration appliance is connected with a water filter provider using a wired connection or a wireless connection to connect the refrigeration appliance with a server via a global communication network of computer servers and wherein status information is provided to the microprocessor based control within the refrigeration appliance and thereafter to the user via a mobile computing device accessed by the user.
